Chunyu Zhao

Department of Economics
University of Macau
Taipa, Macau SAR, China

(+86)13964461560
zhaochunyu.chloe@gmail.com
https://chunyuzhao-chloe.github.io

Education

Ph.D. in Economics, University of Macau	2023 - 2027 (expected)
M.A. in Economics, The University of British Columbia	2022 - 2023
Honours B.A. in Mathematical Economics (Dean's Honours), University of Waterloo	2020 - 2022
BEc. in Financial Mathematics, Shandong University of Finance and Economics	2018 - 2022

Research Interests

Market Design, Matching Theory, Experimental Economics

Working Papers

Mitigating Inequality and Manipulation in Multi-Object Allocation with Inácio Bó
(Presented at the Joint Economics Workshop between the University of Macau and the City University of Macau)

Abstract: In multi-object assignment problems without monetary transfers—such as course allocation and task assignments—sequential mechanisms generate a tension between incentives and fairness. Allowing agents to select several objects in priority order is strategy-proof but can concentrate too much value to those who choose first. Distributing choices across rounds reduces this advantage, but predictable future turns create opportunities for manipulation. We introduce the Switching Random Priority mechanism (SRP), which randomly selects each successive picker among agents other than the previous picker. This prevents consecutive choices while making it risky to postpone selecting a desirable object. We show that, with at least three agents, SRP is weakly SD-strategy-proof and weakly SD-envy-free. Moreover, every realized allocation satisfies ordinal rank balance, a fairness property limiting the concentration of highly ranked objects that quota-based serial dictatorship does not generally satisfy. With two agents, SRP instead becomes SD-efficient but fails weak SD-strategy-proofness.

Backward Elicitation of Preferences in Strategy-Proof Mechanisms with Inácio Bó and Lawrence Choo
(Presented at The Fourth Wuhan Cherry Blossom Workshop on Experimental Economics)

Abstract: We study whether changing preference-elicitation order affects behavior in strategy-proof matching mechanisms. In a laboratory school choice setting, we vary the matching rule (Serial Dictatorship vs. Random Serial Dictatorship) and the ranking procedure (standard vs. backward, eliciting least-preferred schools first). Backward elicitation yields mixed effects: it decreases truthfulness under Random Serial Dictatorship but increases it under Serial Dictatorship. Analysis of decision-making shows participants respond to both payoff differences and perceived school competitiveness, but backward elicitation does not significantly alter sensitivity to either factor.

Papers in Progress

The Informational Value of Historical Cutoffs in Centralized Matching with Inácio Bó and Yubo Tao

Strategy-Proof and Efficient Multi-Object Allocation with Inácio Bó

Teaching and Research Experience

University of Macau

Intermediate Microeconomics (Undergraduate) — Fall 2026
Teaching assistant, with Prof. Ernest Lai

Advanced Microeconomics (Undergraduate) — Spring 2024, 2026
Teaching assistant, with Prof. Inácio Bó

Math and Econ Camp (Ph.D) — Summer 2025
Instructor

Experimental Economics Laboratory — Spring, Fall 2025
Research assistant, with Prof. Lawrence Choo

Intermediate Microeconomics (Undergraduate) — Fall 2024
Teaching assistant, with Prof. Gary Wong

Public Economics (Masters) — Spring 2024
Teaching assistant, with Prof. Guangzhen Sun

The University of British Columbia

Principles of Macroeconomics (Undergraduate) — Spring 2023
Teaching assistant, with Prof. Chowdhury Mahmoud

Making Sense of Economic Data (Undergraduate) — Fall 2022
Teaching assistant, with Prof. Jonathan Graves

Fellowships

UM PhD Assistantship (University of Macau) — 2023–2027

Software

Python, R, Stata, oTree/z-Tree, LaTeX

Languages

Mandarin (native), English (fluent)

References

Prof. Inácio Bó
University of Macau
Department of Economics
Taipa, Macau SAR, China
inaciobo@um.edu.mo

Prof. Lawrence Choo
University of Macau
Department of Economics
Taipa, Macau SAR, China
lawrence.cy.choo@gmail.com

Prof. Yubo Tao
University of Macau
Department of Economics
Taipa, Macau SAR, China
yubotao@um.edu.mo